

First Order Passive Filter w/ Time Constant

Description

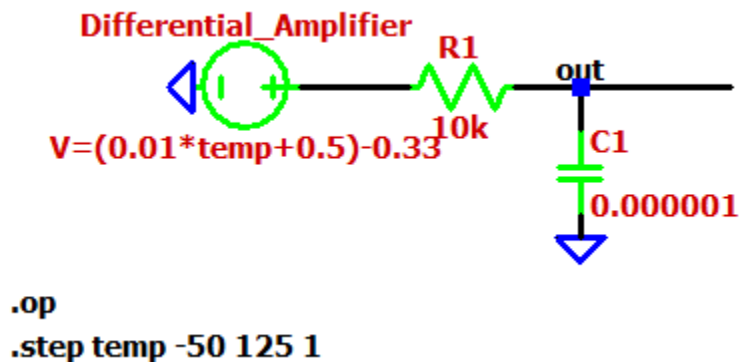
What building block are you applying this concept to?

A first-order passive RC low-pass filter is added after the amplifier to smooth the temperature signal before it is sent to the comparator.

What circuit variable(s) will be derived in analysis? Briefly (1 or 2 sentences) describe how you will apply the concept to derive the variable.

The circuit variables derived are the cutoff frequency and the output voltage of the filter. The cutoff frequency is calculated using the RC time constant, which determines how the filter attenuates high-frequency components of the signal.

Paste a well-formatted schematic of the circuit you are analyzing, and include only this circuit and not any others. Label all important nodes; Any nodes or components you discuss in your analysis section should correspond to a labelled node in the schematic.



Analysis

Write the general, symbolic equation(s) you are using to perform your calculation. Please show any lengthy derivations in the appendix to improve readability. Ensure that all

variables used in the equation are defined either by labelling them in the schematic or defining them in-text.

$$F_c = 1/(2\pi RC)$$

Calculate a numerical value for your circuit variable(s).

Using $R = 10\text{k}\Omega$ and $C = 1\mu\text{F}$:

$$F_c = 15.9\text{ Hz}$$

State what circuit variable(s) you will measure in your simulation and experimental section to compare with your analysis and confirm intended operation of your circuit.

The output voltage of the filter to verify that the signal is smoother and noise is reduced.

Simulation

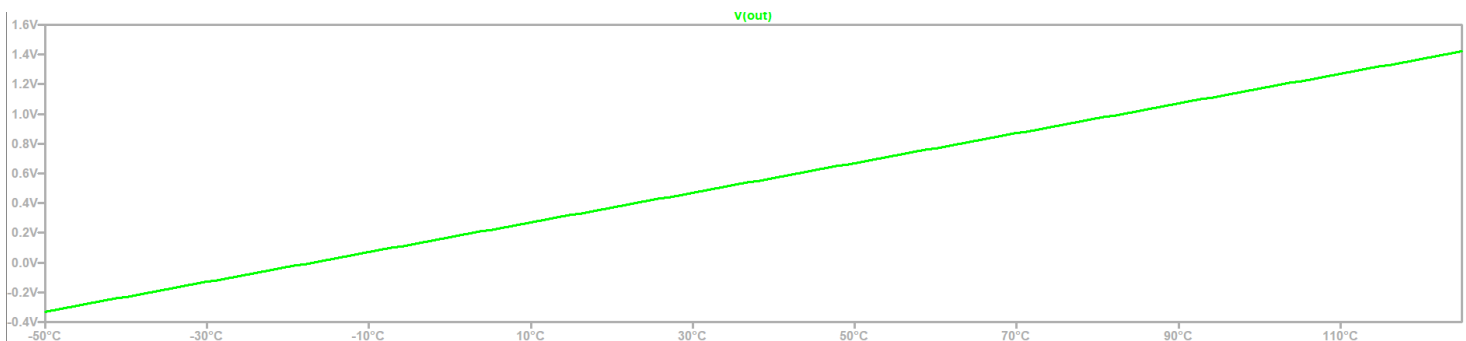
Describe your simulation in 1 to 2 sentences.

A DC sweep simulation was performed using `.step temp -50 125 1` to observe how the first-order RC passive filter responds to the temperature-dependent input signal. The output voltage was measured at the filter node to evaluate how the filter smooths the signal as temperature varies.

If your simulation schematic differs from your schematic in the description, include it here. Make sure you include all simulation commands (e.g., `'tran 1m.'`).

Our simulation schematic differs slightly from the schematic in the description. Specifically, when simulating only the first-order passive filter, the differential amplifier (OP484) was replaced with an equivalent behavioral source to simplify the analysis. As a result, the simulation does not capture the output saturation behavior of the op-amp, where V_{out} is limited between 0V and 5V depending on the difference between V_{in+} and V_{in-} . Instead, the purpose of this simulation is to isolate and demonstrate the behavior of the first-order RC passive filter in response to the temperature-dependent input signal.

Place your simulation results here. This can be a plot, a numerical value, or a table. Make sure your data is formatted well and easy to read. Include all axes on plots and make sure it is clear what value is being measured.

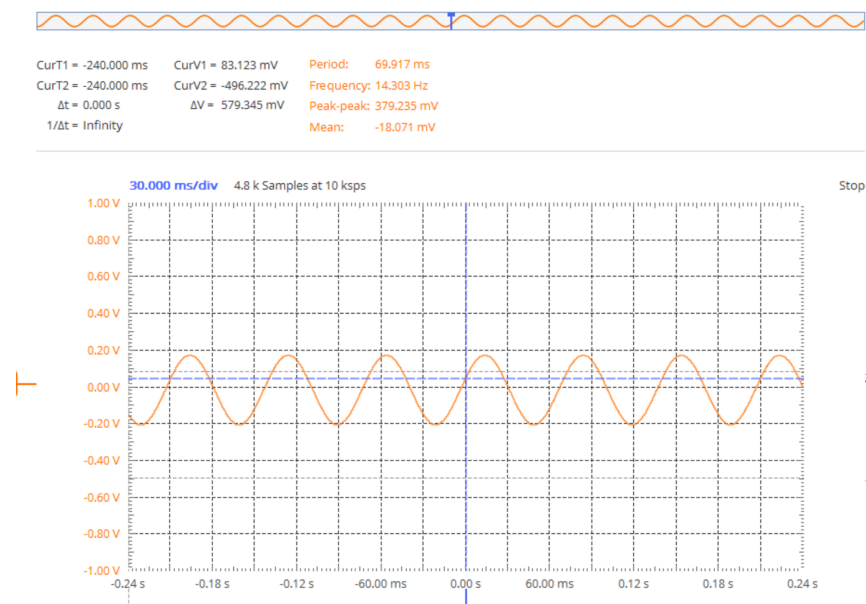


Experimental

Describe your measurement in 1 to 3 sentences. Describe any differences between your analysis/simulation and your experimental circuit. List the part numbers of any components you used outside of passive components (R,L,C), for example which op amp you used.

A sinusoidal voltage source was applied to the input of the RC low-pass filter, and the input and output signals were measured using the oscilloscope (CH1 at the input and CH2 at the output). The frequency was adjusted until the output amplitude was approximately 70.7% of the input amplitude, corresponding to the -3 dB cutoff condition for a first-order filter. This occurred at approximately 14.3 Hz, which is close to the calculated cutoff frequency of 15.9 Hz. The difference is attributed to component tolerances and non-ideal effects of the OP484.

Place your simulation results here. This can be a plot, a numerical value, or a table. Make sure your data is formatted well and easy to read. Include all axes on plots and make sure it is clear what value is being measured.



Discussion

The analytical and simulation results closely matched, with a cutoff frequency around 15.9 Hz and a smoother output signal. Experimental results also showed reduced noise, confirming

proper filter operation. Minor differences are due to component tolerances and loading effects from the next stage. The percent error for this is 10.1%.

Discuss how this analysis helped you in your design.

This analysis helped ensure a stable, noise-free signal for the rest of the circuit. This ensures that high-frequency noise is removed before later stages of the circuit.

Non-inverting Amplifier

Description

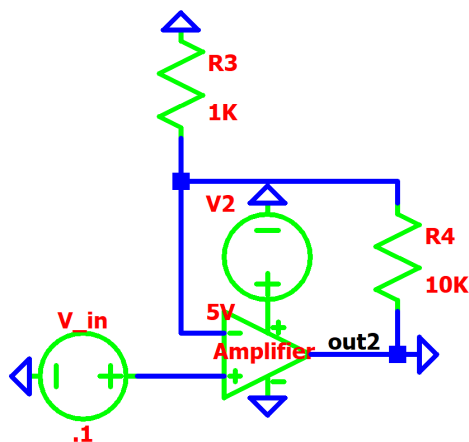
What building block are you applying this concept to?

A non-inverting amplifier is used to increase the small voltage output from the TMP36 temperature sensor so it can be processed more effectively.

What circuit variable(s) will be derived in analysis? Briefly (1 or 2 sentences) describe how you will apply the concept to derive the variable.

The circuit variables derived are the voltage gain and the output voltage of the amplifier. The gain is determined using the resistor ratio in the non-inverting amplifier configuration, which is then used to calculate the amplified output signal.

Paste a well-formatted schematic of the circuit you are analyzing, and include only this circuit and not any others. Label all important nodes; Any nodes or components you discuss in your analysis section should correspond to a labelled node in the schematic.



Analysis

Write the general, symbolic equation(s) you are using to perform your calculation. Please show any lengthy derivations in the appendix to improve readability. Ensure that all variables used in the equation are defined either by labelling them in the schematic or defining them in-text.

$$V_{out} = (1 + R_4/R_3) \cdot V_{in+}$$

Calculate a numerical value for your circuit variable(s).

$$V_{out} = (1 + 10K/1K) \cdot (0.35V) = 3.85V$$

State what circuit variable(s) you will measure in your simulation and experimental section to compare with your analysis and confirm intended operation of your circuit.

The input and output voltage of the amplifier to confirm the expected gain.

Simulation

Describe your simulation in 1 to 2 sentences.

In this simulation we have a non-inverting amplifier that will amplify the voltage going into the positive terminal by 11x. In this case we chose 0.1V as the V_{in+}

If your simulation schematic differs from your schematic in the description, include it here. Make sure you include all simulation commands (e.g., '.tran 1m.').

For the sake of the simulation for this block, we attached a 0.1V powersource to V_{in+} . Realistically, V_{in+} would be the scaling and filtered voltage from the previous two blocks. In this case we wanted to demonstrate how the non-inverting amplifier would amplify the given voltage by 11x.

Place your simulation results here. This can be a plot, a numerical value, or a table. Make sure your data is formatted well and easy to read. Include all axes on plots and make sure it is clear what value is being measured.

```

C:\Users\Carter\Downloads\Draft2.net

--- Operating Point ---

V(n001):      4.54545e-06  voltage
V(n002):      5           voltage
V(n003):      0.1         voltage
I(R3):        -4.54545e-09 device_current
I(R4):        4.54545e-10 device_current
I(V2):        -0.025      device_current
I(V_in):      4.8e-09     device_current
Ix(u1:1):     -4.8e-09    subckt_current
Ix(u1:2):     -4.99999e-09 subckt_current
Ix(u1:3):     0.025       subckt_current
Ix(u1:4):     -7.18605e-08 subckt_current
Ix(u1:5):     -0.025      subckt_current

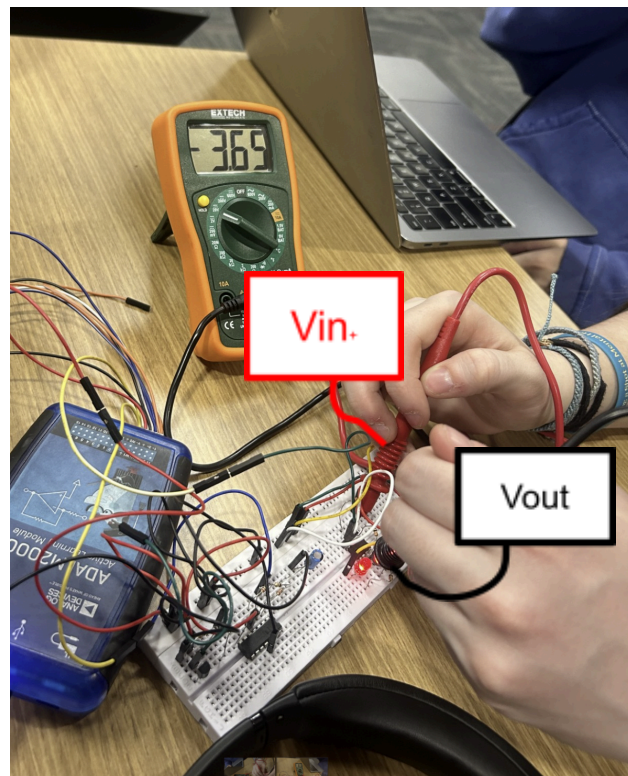
```

Experimental

Describe your measurement in 1 to 3 sentences. Describe any differences between your analysis/simulation and your experimental circuit. List the part numbers of any components you used outside of passive components (R,L,C), for example which op amp you used.

For our experiment instead of using the temperature sensor as our voltage input, a 0.35 V DC input was applied to the circuit. The OP97 is the core part of the non-inverting amplifier.

Place your simulation results here. This can be a plot, a numerical value, or a table. Make sure your data is formatted well and easy to read. Include all axes on plots and make sure it is clear what value is being measured.



Discussion

Compare your analytical, simulation, and experimental results. Discuss any differences between them. If there is any difference between calc, sim, and measurement provide a reason as to why (DO NOT say it is from 'human error' or 'incorrect wiring').

The calculated gain of 11 matched well with simulation and experimental results, confirming correct amplifier behavior. Small differences are likely due to op-amp non-idealities and resistor tolerances. The percent error between analytic and experimental using the values 3.85 as accepted and 3.65 as measured is 5.2%.

Discuss how this analysis helped you in your design.

This stage was important for boosting the small sensor voltage to a usable level for further processing.

First-Order Passive Filter w/ Thevenin Equivalent

Description

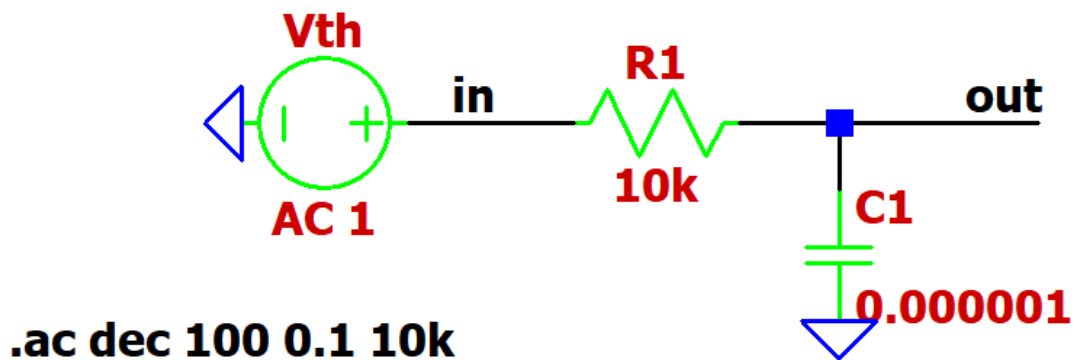
What building block are you applying this concept to?

The Thevenin equivalent is applied to the circuitry preceding the first-order RC filter by replacing the differential amplifier stage with an equivalent voltage source and resistance at the input node of the filter.

What circuit variable(s) will be derived in analysis? Briefly (1 or 2 sentences) describe how you will apply the concept to derive the variable.

The circuit variables derived are the Thevenin voltage (V_{th}) and Thevenin resistance (R_{th}). These are obtained by analyzing the differential amplifier output and determining the equivalent voltage and resistance seen at the input of the RC filter.

Paste a well-formatted schematic of the circuit you are analyzing, and include only this circuit and not any others. Label all important nodes; Any nodes or components you discuss in your analysis section should correspond to a labelled node in the schematic.



Analysis

Write the general, symbolic equation(s) you are using to perform your calculation. Please show any lengthy derivations in the appendix to improve readability. Ensure that all variables used in the equation are defined either by labelling them in the schematic or defining them in-text.

$$V_{th} = V_{out}(\text{differential amplifier}) = V_{in}(\text{filter})$$

$$V_{out} = (R_8/R_7)(V_{in+} - V_{in-})$$

$$\text{Given } R_8 = R_7$$

$$V_{th} = V_{in+} - V_{in-}$$

$$R_{th} = V_{test}/I_{test}$$

Calculate a numerical value for your circuit variable(s).

Assuming we use a V_{in-} of 0.1V and the temperature sensor outputs a V_{in+} of 1V,

$$V_{th} = 1 - 0.1 = 0.9V$$

$$\mathbf{V_{th} = V_{out} = 0.9V}$$

To determine the Thevenin resistance, the resistance seen looking into the output of the op-amp is evaluated. For an ideal op-amp, the output behaves as an ideal voltage source with zero output resistance, meaning it can maintain its output voltage regardless of the load. Therefore,

$$\mathbf{R_{th} = 0}$$

Since $R_{th} = 0$, the filter isn't affected by the previous stages in the circuit, meaning the filter behaves exactly like an ideal first-order filter with:

$$\text{Time constant} = RC,$$

$$F_c = 1/(2\pi RC)$$

State what circuit variable(s) you will measure in your simulation and experimental section to compare with your analysis and confirm intended operation of your circuit.

The voltage at the input of the RC filter (output of the differential amplifier) will be measured and compared to the calculated Thevenin voltage. Additionally, consistency of this voltage under loading will confirm that the Thevenin resistance is approximately zero.

Simulation

Describe your simulation in 1 to 2 sentences.

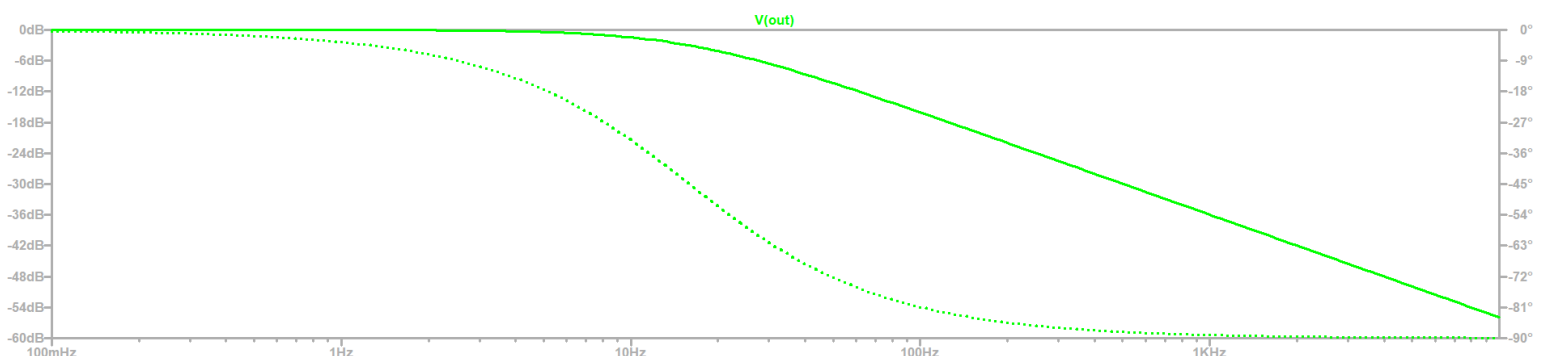
An AC sweep simulation was performed to analyze the frequency response of the first-order RC low-pass filter. The differential amplifier stage was replaced by its Thevenin equivalent, modeled as an ideal AC voltage source with magnitude 1 V, to observe the filter's gain and cutoff behavior.

If your simulation schematic differs from your schematic in the description, include it here. Make sure you include all simulation commands (e.g., '.tran 1m.').

The simulation schematic differs from the original circuit in that the differential amplifier was replaced with its Thevenin equivalent. Specifically, the amplifier output was modeled as an ideal voltage source (V_{th}) with AC magnitude of 1 V and zero output resistance, allowing the RC filter to be analyzed independently. The simulation used the command (.ac dec 100 0.1 10k) to sweep frequencies from 0.1 Hz to 10 kHz.

Place your simulation results here. This can be a plot, a numerical value, or a table. Make sure your data is formatted well and easy to read. Include all axes on plots and make sure it is clear what value is being measured.

The AC sweep results show that the output voltage remains approximately constant at low frequencies and begins to attenuate near the cutoff frequency of about 15.9 Hz. Beyond this point, the magnitude decreases at a rate of approximately -20 dB/decade, confirming the expected first-order low-pass filter behavior.



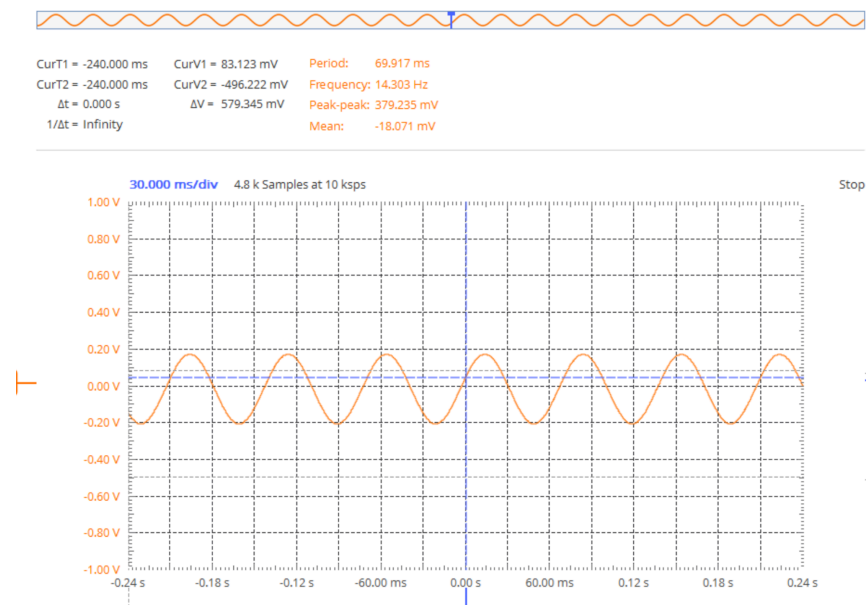
Experimental

Describe your measurement in 1 to 3 sentences. Describe any differences between your analysis/simulation and your experimental circuit. List the part numbers of any components you used outside of passive components (R,L,C), for example which op amp you used.

The experimental measurement was performed by applying a sinusoidal input signal to the RC filter and observing the output voltage across the capacitor using an oscilloscope. The measured cutoff frequency was approximately 14.3 Hz, which is close to the calculated value of 15.9 Hz. This confirms that the filter behaves as expected in practice.

The experimental circuit includes the output of the differential amplifier rather than an ideal Thevenin source; however, because the OP484 has a very low output resistance, it behaves similarly to an ideal voltage source. As a result, the filter response closely matches the theoretical and simulated results, with small deviations due to component tolerances and non-ideal op-amp behavior.

Place your simulation results here. This can be a plot, a numerical value, or a table. Make sure your data is formatted well and easy to read. Include all axes on plots and make sure it is clear what value is being measured.



Discussion

Compare your analytical, simulation, and experimental results. Discuss any differences between them. If there is any difference between calc, sim, and measurement provide a reason as to why (DO NOT say it is from 'human error' or 'incorrect wiring').

The analytical, simulation, and experimental results all showed consistent low-pass filter behavior, with a cutoff frequency near the expected value of 15.9 Hz. The simulation closely matched the analytical prediction, while the experimental cutoff frequency was slightly lower (approximately 14.3 Hz). This small difference is primarily due to component tolerances in the resistor and capacitor values, as well as non-ideal effects such as the finite bandwidth and non-zero output impedance of the op-amp, which slightly alter the effective time constant of the circuit.

Discuss how this analysis helped you in your design.

The Thevenin equivalent analysis helped confirm that the differential amplifier stage does not significantly affect the filter's behavior, since the op-amp presents a very low output resistance. This allowed the RC filter to be designed independently using the standard cutoff frequency formula, simplifying the overall design process and ensuring predictable performance when integrating multiple circuit stages.

First-Order Circuit w/ s-Domain Analysis

Description

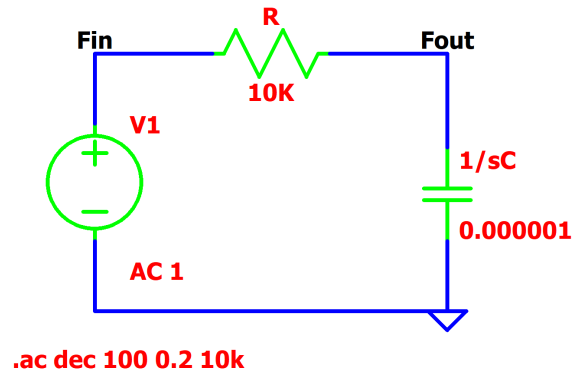
What building block are you applying this concept to?

We are applying this concept to our first-order passive filter by analyzing it in the s-Domain.

What circuit variable(s) will be derived in analysis? Briefly (1 or 2 sentences) describe how you will apply the concept to derive the variable.

The circuit variables derived are the cutoff frequency and the output voltage of the filter. The cutoff frequency is calculated using s-domain analysis to find the cutoff where higher frequencies will be filtered.

Paste a well-formatted schematic of the circuit you are analyzing, and include only this circuit and not any others. Label all important nodes; Any nodes or components you discuss in your analysis section should correspond to a labelled node in the schematic.



Analysis

Write the general, symbolic equation(s) you are using to perform your calculation. Please show any lengthy derivations in the appendix to improve readability. Ensure that all variables used in the equation are defined either by labelling them in the schematic or defining them in-text.

$$H(s) = \frac{V_{out}}{V} = \frac{\frac{1}{sC}}{R + \frac{1}{sC}} = \frac{1}{sCR + 1}$$

$$s = j\omega \rightarrow H(j\omega) = \frac{1}{1 + j\omega CR}$$
 Cutoff frequency happens when the real and complex parts are equal:

$$\omega CR = 1 \rightarrow \omega = \frac{1}{CR}$$

$$\omega = 2\pi f \rightarrow f = \frac{\omega}{2\pi} \rightarrow \underline{f_c = \frac{1}{2\pi CR}}$$

Calculate a numerical value for your circuit variable(s).

Using $R = 10\text{k}\Omega$ and $C = 1\mu\text{F}$:

$$f_c = 1/(2\pi \cdot 10\text{k} \cdot 1\mu) = 15.9 \text{ Hz}$$

State what circuit variable(s) you will measure in your simulation and experimental section to compare with your analysis and confirm intended operation of your circuit.

We will measure the cutoff frequency of the first-order passive filter.

Simulation

Describe your simulation in 1 to 2 sentences.

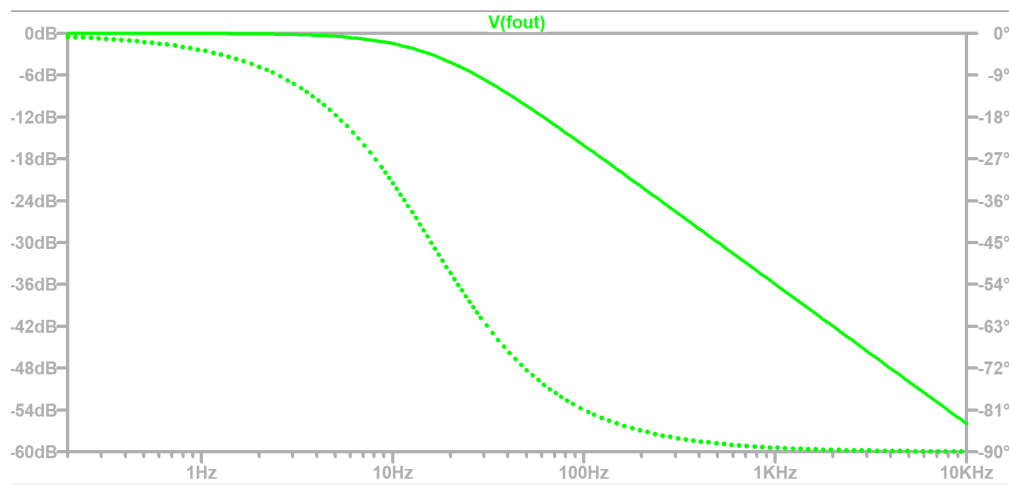
An AC sweep simulation was performed to analyze the frequency response of the first-order RC low-pass filter. A 1V AC voltage source was supplied and went into the resistor and the capacitor, which are in series.

If your simulation schematic differs from your schematic in the description, include it here. Make sure you include all simulation commands (e.g., '.tran 1m.').

The simulation schematic differs from the original circuit in that it just tests the single block; a 1V AC voltage source was put in as Vin. This allows the RC filter to be analyzed independently. The simulation used the command (.ac dec 100 0.1 10k) to sweep frequencies from 0.1 Hz to 10 kHz.

Place your simulation results here. This can be a plot, a numerical value, or a table. Make sure your data is formatted well and easy to read. Include all axes on plots and make sure it is clear what value is being measured.

The AC sweep results show that the output voltage remains approximately constant at low frequencies and begins to attenuate near the cutoff frequency of about 15.9 Hz. Beyond this point, the magnitude decreases at a rate of approximately -20 dB/decade, confirming the expected first-order low-pass filter behavior.

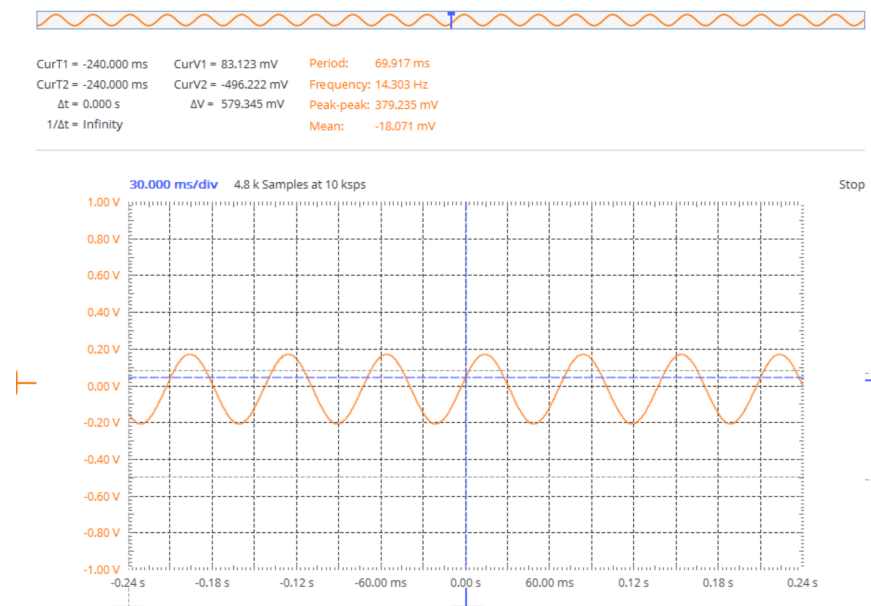


Experimental

Describe your measurement in 1 to 3 sentences. Describe any differences between your analysis/simulation and your experimental circuit. List the part numbers of any components you used outside of passive components (R,L,C), for example which op amp you used.

A sinusoidal voltage source was applied to the input of the RC low-pass filter, and the input and output signals were measured using the oscilloscope (CH1 at the input and CH2 at the output). The frequency was adjusted until the output amplitude was approximately 70.7% of the input amplitude, corresponding to the -3 dB cutoff condition for a first-order filter. This occurred at approximately 14.3 Hz, which is close to the calculated cutoff frequency of 15.9 Hz. The difference is attributed to component tolerances and non-ideal effects of the OP484.

Place your simulation results here. This can be a plot, a numerical value, or a table. Make sure your data is formatted well and easy to read. Include all axes on plots and make sure it is clear what value is being measured.



Discussion

Compare your analytical, simulation, and experimental results. Discuss any differences between them. If there is any difference between calc, sim, and measurement provide a reason as to why (DO NOT say it is from 'human error' or 'incorrect wiring').

The analytical and simulation results closely matched, with a cutoff frequency around 15.9 Hz and a smoother output signal. Experimental results also showed reduced noise, confirming proper filter operation. Minor differences are due to component tolerances and loading effects from the next stage. The percent error for this is -10.1%.

Discuss how this analysis helped you in your design.

The s-Domain analysis helped confirm that the low-pass filter would effectively attenuate high-frequency switching noise from the differential amplifier. By deriving the transfer function, the design was optimized to place the pole at a frequency that balances signal integrity with the system's reaction time.